THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The legs of a right triangle are 5 cm and 7 cm. Its hypotenuse (in cm) is: 1

(a) $\sqrt{74}$ **(b)** $2\sqrt{3}$ **(c)** 12 **(d)** $5\sqrt{3}$

2. The legs of a right triangle are 7 cm and 9 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 256 **(b)** 63 **(c)** 16 **(d)** 130

3. Taking $m = 8$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 16, 63, 66 **(b)** 16, 64, 65 **(c)** 8, 63, 65 **(d)** 16, 63, 65

4. A triangle has sides 14 cm, 48 cm and 50 cm. It is: 1

(a) equilateral **(b)** not right-angled **(c)** cannot be said **(d)** right-angled

5. The diagonal of a square of side 3 cm (in cm) is: 1

(a) 6 **(b)** 3 **(c)** $3\sqrt{2}$ **(d)** $\sqrt{19}$

6. The hypotenuse of a right triangle is 29 cm and one leg is 20 cm. Its area (in cm²) is: 1

(a) 210 **(b)** 290 **(c)** 230 **(d)** 420

7. The Baudhayana–Pythagoras theorem states that in a right triangle: 1

(a) the square of a leg equals the sum of the other two sides
(b) the square of the hypotenuse equals the sum of the squares of the other two sides
(c) the hypotenuse is the shortest side
(d) the sum of the two legs equals the hypotenuse

8. The legs of a right triangle are 2 cm and 6 cm. Its hypotenuse (in cm) is: 1

(a) $2\sqrt{10}$ **(b)** 8 **(c)** $2\sqrt{2}$ **(d)** $\sqrt{41}$

9. The legs of a right triangle are 11 cm and 13 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 290 **(b)** 576 **(c)** 143 **(d)** 24

10. Taking $m = 9$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 18, 81, 82 **(b)** 18, 80, 82 **(c)** 18, 80, 83 **(d)** 9, 80, 82

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the Pythagorean triplet with the smallest member 28. 2

12. The sides of a triangle are 16 cm, 63 cm, 65 cm. Is it a right triangle? Which angle is the right angle? 2

13. Find the length of the diagonal of a square of side 4 cm. 2

14. Find the Pythagorean triplet with the smallest member 30. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The hypotenuse of an isosceles right triangle is $4\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

16. A ladder 29 m long reaches a window 21 m above the ground. Find the distance of the foot of the ladder from the wall. 3

17. The diagonals of a rhombus are 20 cm and 48 cm. Find its side. 3

OR

The hypotenuse of an isosceles right triangle is $6\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

18. A ladder 41 m long reaches a window 40 m above the ground. Find the distance of the foot of the ladder from the wall. 3

OR

The diagonals of a rhombus are 24 cm and 70 cm. Find its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The diagonals of a rhombus are 40 cm and 42 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 20 m and 12 m stand on level ground 15 m apart. Find the distance between their tops. 5

20. The diagonals of a rhombus are 18 cm and 80 cm. Find its side, perimeter and area. 5

OR

Two poles of heights 30 m and 6 m stand on level ground 10 m apart. Find the distance between their tops. 5

END OF QUESTION PAPER — Total: 40 Marks

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